



MISO DPP 2019 Central Area Study Phase II Report

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Revision 03

MISO
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1. Executive Summary

This report presents the results of a System Impact Study (SIS) performed to evaluate the interconnection of the generators in the DPP 2019 Central Area Phase II Study Cycle. The study was performed under the direction of MISO and reviewed by an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Ameren (AMIL, AMMO, ATXI), Duke Energy (DEI), Hoosier Energy (HE), Indianapolis Power & Light Company (IPL), Northern Indiana Public Service Company (NIPSCO), Southern Illinois Power Cooperative (SIPC), Big Rivers Electric Corporation (BREC), Henderson Municipal Power & Light (HMP&L), Columbia Water & Light (CWLD), City Water Light & Power (CWLP) and Southern Indiana Gas & Electric Company (SIGE).

1.1. Project List

The original interconnection requests for the DPP 2019 Central Area Study Cycle had a total of 110 generation projects.

- Projects J1240, J1273, J1418, J1420, J1412, J1288, J1324, J1489 withdrew prior to the Phase I study. This left 102 projects being active for the DPP Phase I study process.
- 102 generation projects with a combined nameplate rating of 15,725.6 MW (ERIS) & 15,515.6 MW (NRIS) were studied in Phase I.
- The 2019 DPP Phase I Central Area study was kicked off on June 23rd 2020

The 2019 DPP Phase II Central Area study consisted of the following:

- Decision Point 1 (DP1): 21 Projects withdrew
 - J1197, J1222, J1225, J1227, J1242, J1243, J1254, J1271, J1274, J1275, J1296, J1301, J1318, J1330, J1338, J1363, J1376, J1385, J1398, J1409, J1457
- Total projects remaining for phase II Study: 81 (12,766.7 MW of ERIS & 12,332 MW of NRIS)

Table 1: List of 2019 DPP Phase II Central Area Projects

Project #	Fuel	Transmission Owner	County	State	Service Type	Decision Point 1 ERIS MW	Decision Point 1 NRIS MW	POI Name
J1191	Solar	City of Columbia, MO	Boone County	MO	NRIS	64	64	Bolstad 69 kV Substation
J1198	Hybrid	Ameren Illinois	Jasper County	IL	NRIS	51.8	51.8	Newton 345kV Substation
J1199	Hybrid	Ameren Illinois	Fulton County	IL	NRIS	20	20	Duck Creek 345kV Substation
J1200	Hybrid	Ameren Illinois	Putnam County	IL	NRIS	24.2	24.2	Putnam 138kV Substation
J1201	Hybrid	Ameren Illinois	Montgomery County	IL	NRIS	44.2	44.2	North Coffeen 138kV Substation
J1202	Hybrid	Ameren Illinois	Randolph County	IL	NRIS	68.4	68.4	Baldwin Plant 345kV Switchyard - Baldwin Unit 2
J1204	Solar	Ameren Illinois	Cumberland County	IL	NRIS	199.3	199.3	Casey West 345kV Substation

J1208	Solar	Ameren Illinois	Saline County	IL	NRIS	80	80	Muddy - Crab Orchard 138 kV line
J1209	Solar	Ameren Illinois	Saline County	IL	NRIS	80	80	Muddy - Crab Orchard 138kV Line
J1213	Hydro	Ameren Missouri	Iron County	MO	NRIS	60	60	Taum Sauk 138kV Switchyard, Bus 1 & 2
J1216	Solar	Ameren Illinois	Washington County	IL	NRIS	185	185	Ashley 138 kV Substation
J1231	Solar	Big Rivers Electric Corporation	Lyon County	KY	NRIS	125	125	Barkley - Caldwell 161kV Line
J1232	Solar	Ameren Illinois	Coles County	IL	NRIS	99	99	Chesterville - Mattoon West 138kV Line
J1233	Battery Storage	Southern Indiana Gas & Electric Company d/b/a Vectren Energy Delivery of Indiana, Inc.	Pike County	IN	NRIS	32.5	32.5	Cato 138 kV Substation
J1234	Solar	Duke Energy Indiana	Knox County	IN	NRIS	100	100	Edwardsport - Washington Mun, Linton 138kV Line
J1235	Solar	Duke Energy Indiana	Knox County	IN	NRIS	50	50	Edwardsport - Washington Mun, Linton 138kV Line
J1241	Solar	Ameren Illinois	Jefferson County	IL	NRIS	165	165	West Mt Vernon - Xenia 345kV Line
J1263	Hybrid	Ameren Illinois	Clark County	IL	NRIS	408.6	408.6	Casey - Kansas West 345kV Line
J1265	Solar	Northern Indiana Public Service Company	LaGrange County	IN	NRIS	43	43	Wolcottville - South Milford 69kV Line
J1266	Solar	Ameren Illinois	Marion County	IL	NRIS	99	99	Kinmundy - Salem 138kV Line
J1268	Solar	Ameren Missouri	Lincoln County	MO	NRIS	150	88	Pike - Troy 161kV Line
J1269	Hybrid	Southern Indiana Gas & Electric Company d/b/a Vectren Energy Delivery of Indiana, Inc.	Warrick County	IN	NRIS	100	47.6	Elliott 69kV Substation
J1272	Battery Storage	Southern Indiana Gas & Electric Company d/b/a Vectren Energy Delivery of Indiana, Inc.	Vanderburgh County	IN	NRIS	50	23.6	Angel Mound 138kV Substation
J1276	Hybrid	Southern Indiana Gas & Electric Company d/b/a Vectren Energy Delivery of Indiana, Inc.	Posey County	IN	NRIS	100	47.1	Point 69kV Substation

J1289	Wind	Ameren Illinois	Morgan County	IL	NRIS	200	200	Turner - Austin 345kV Line
J1295	Solar	Duke Energy Indiana	Gibson County	IN	NRIS	280	280	Gibson - Francisco 345kV Line
J1299	Solar	Ameren Missouri	Scott County	MO	NRIS	149	149	Miner - Kelso 161kV Line
J1302	Solar	Ameren Illinois	Saline County	IL	NRIS	150	150	Muddy - Norris City 138kV Line
J1303	Solar	Southern Illinois Power Cooperative	Jackson County	IL	NRIS	95	95	Campbell Hill - Jackson 161kV Line
J1306	Solar	Ameren Illinois	Perry County	IL	NRIS	200	200	Commodore-Jordan 345kV Line
J1308	Solar	Southern Indiana Gas & Electric Company d/b/a Vectren Energy Delivery of Indiana, Inc.	Posey County	IN	NRIS	300	300	AB Brown - Gibson 345kV Line
J1311	Solar	Ameren Illinois	St. Clair County	IL	NRIS	150	150	Fayetteville Bee Hollow Road 138 kV Substation
J1323	Battery Storage	Northern Indiana Public Service Company	LaGrange County	IN	NRIS	25	25	Lagrange 138 kV Substation
J1332	Solar	Northern Indiana Public Service Company	Jasper County	IN	NRIS	200	200	Schahfer 345kV Substation
J1333	Hybrid	Northern Indiana Public Service Company	Jasper County,Starke County	IN	NRIS	199.5	199.5	Schahfer - Burr Oak 345kV Line
J1334	Hybrid	Northern Indiana Public Service Company	Jasper County,Starke County	IN	NRIS	199.5	199.5	Schahfer - Burr Oak 345kV Line
J1335	Hybrid	Northern Indiana Public Service Company	Jasper County,Starke County	IN	NRIS	199.5	199.5	Schahfer - Burr Oak 345kV Line
J1336	Hybrid	Northern Indiana Public Service Company	Jasper County,Starke County	IN	NRIS	199.5	199.5	Schahfer - Burr Oak 345kV Line
J1339	Hybrid	Northern Indiana Public Service Company	Jasper County,Starke County	IN	NRIS	199.5	199.5	Schahfer - Burr Oak 345kV Line
J1340	Hybrid	Northern Indiana Public Service Company	Jasper County,Starke County	IN	NRIS	199.5	199.5	Schahfer - Burr Oak 345kV Line
J1342	Solar	Northern Indiana Public Service Company	Starke County	IN	NRIS	99	99	Starke 138kV Substation
J1348	Solar	Duke Energy Indiana	Vermillion County	IN	NRIS	200	200	Cayuga - Sugar Creek 345kV Line

J1352	Solar	Ameren Missouri	Montgomery County	MO	NRIS	100	100	Spencer Creek - Montgomery 345kV Line
J1353	Battery Storage	Northern Indiana Public Service Company	Lake County	IN	NRIS	150	150	Green Acres 138kV Substation
J1354	Battery Storage	Hoosier Energy	Decatur County	IN	NRIS	118	118	Decatur County 138kV Switching Station
J1355	Battery Storage	Northern Indiana Public Service Company	LaPorte County	IN	NRIS	131	131	Stillwell 138kV Substation
J1356	Solar	Duke Energy Indiana	Martin County	IN	NRIS	135	74	Whitfield 138kV Substation
J1357	Battery Storage	Duke Energy Indiana	Madison County	IN	NRIS	150	150	Madison 138 KV SUBSTATION
J1358	Battery Storage	Northern Indiana Public Service Company	LaPorte County	IN	NRIS	125	125	Luchtman 138kV Substation
J1360	Wind	Ameren Illinois	Piatt County	IL	NRIS	300	300	Goose Creek 345kV Switchyard
J1369	Solar	Southern Indiana Gas & Electric Company d/b/a Vectren Energy Delivery of Indiana, Inc.	Dubois County	IN	NRIS	100	100	Dubois - Culley 138 kV Line
J1378	Solar	Duke Energy Indiana	Fountain County	IN	NRIS	200	200	Veedersburg West 230 kV Substation
J1381	Solar	Duke Energy Indiana	Howard County	IN	NRIS	200	200	Greentown 230 kV substation
J1382	Solar	Ameren Illinois	Clark County	IL	NRIS	150	150	Hutsonville - Neoga 138 kV Line tap
J1383	Solar	Ameren Illinois	Fulton County	IL	NRIS	150	150	Ipava 138 kV Substation
J1386	Solar	Northern Indiana Public Service Company	Lake County	IN	NRIS	100	100	Green Acres 138 kV Substation
J1387	Solar	Northern Indiana Public Service Company	Jasper County	IN	ERIS	60	0	Circuit 6958 69 kV Line Tap
J1388	Solar	Duke Energy Indiana	Gibson County	IN	ERIS	150	0	Oakland City - Old Ben Coal 138 kV Line Tap
J1390	Solar	Duke Energy Indiana	Bartholomew County	IN	NRIS	150	150	Gallagher Station - Columbus 230 kV Line Tap
J1391	Solar	Southern Indiana Gas & Electric Company d/b/a Vectren Energy Delivery of Indiana, Inc.	Dubois County	IN	NRIS	100	100	Duff 138 kV Substation

J1392	Solar	Northern Indiana Public Service Company	LaGrange County	IN	NRIS	100	100	Lagrange 138 kV Substation
J1393	Solar	Southern Indiana Gas & Electric Company d/b/a Vectren Energy Delivery of Indiana, Inc.	Pike County	IN	NRIS	130	130	Cato 138 kV Substation
J1396	Solar	Hoosier Energy	Pike County	IN	NRIS	60	60	Ratts-Victory 161kV Line
J1405	Solar	Duke Energy Indiana	Greene County	IN	NRIS	140	140	Sanborn - Linton 138 kV Line Tap
J1407	Solar	Northern Indiana Public Service Company	Porter County	IN	NRIS	100	100	South Valparaiso 138 kV Substation
J1408	Hybrid	Ameren Illinois	Macoupin County	IL	NRIS	100	100	Viriden South 138kV Substation
J1419	Solar	Duke Energy Indiana	Greene County	IN	NRIS	250	250	Edwardsport - Amo 345 kV Line Tap
J1422	Hybrid	Ameren Illinois	Edwards County	IL	NRIS	150	150	Albion - Olney 138 kV Line Tap
J1447	Solar	Northern Indiana Public Service Company	Porter County	IN	NRIS	250	250	Schahfer - Green Acres 345kV Line
J1448	Solar	Big Rivers Electric Corporation	Caldwell County	KY	NRIS	150	150	Hopkins - Caldwell County - Barkely 161kV Line
J1450	Solar	Big Rivers Electric Corporation	Webster County	KY	NRIS	150	150	Hopkins - Reid161kV Line
J1453	Wind	Ameren Illinois	Mason County	IL	NRIS	165	165	Havana - Shockey 138 kV Line Tap
J1454	Wind	Ameren Illinois	Mason County	IL	NRIS	40.9	40.9	Mason 138 kV Substation
J1464	Solar	Ameren Transmission Company of Illinois	Macoupin County, Morgan County, Sangamon County	IL	NRIS	592.8	592.8	Meredosia - Austin 345 kV Line tap
J1466	Solar	Big Rivers Electric Corporation	Henderson County	KY	NRIS	250	250	Reid EHV 161 kV Substation
J1470	Wind	Duke Energy Indiana	Benton County, Warren County	IN	NRIS Only	0	30	Cayuga CT 345kV Substation
J1475	Solar	Ameren Illinois	Clark County	IL	NRIS	75	75	Casey West - Sullivan 345 kV Line Tap
J1481	Solar	Duke Energy Indiana	Howard County	IN	NRIS	200	200	Greentown 230 kV substation
J1482	Solar	Northern Indiana Public Service Company	White County	IN	NRIS	150	150	Monticello - Springboro 138 kV Line Tap



J1488	High Voltage DC	Ameren Missouri	Randolph County	MO	NRIS	500	500	McCredie - Montgomery 345 kV Line Tap
J1490	High Voltage DC	Ameren Missouri	Randolph County	MO	NRIS	1000	1000	McCredie - Montgomery 345 kV Line Tap



1.2. Total Network Upgrades

The cost allocation of Network Upgrades for the projects in the DPP 2019 Central Area Phase II Study Cycle reflects responsibilities for mitigating system impacts. The total cost of network upgrades is listed in Table 2 below. The costs for Network Upgrades are planning-level estimates and subject to revision in the facility studies.

Table 2: Total Cost of Network Upgrades for DPP 2019 Central Phase II Projects

Project	ERIS Network Upgrades (\$)					NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Network Upgrade Cost for Milestone Calculation	M2 (\$)	M3 (\$)	M4 (\$)
	Thermal	Voltage	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO – Owned Direct Assigned		($\text{\$}$)			
a	b	c	d	e	f	g	h	i	j	k = b+c+d+e+g+h+j	\$4,000/MW	10% of (k) from Phase I-M2	(20% of (k) from Phase II) – M2 – M3
J1191	1,031,401	0	0	0	2,096,784	0	473,600	0	0	1,505,001	256,000	0	45,000
J1198	423,509	0	0	0	0	0	0	52,000	0	423,509	208,000	0	0
J1199	0	0	0	0	0	0	504,000	854,000	0	504,000	80,000	0	20,800
J1200	58,033	0	0	0	1,230,000	0	532,000	1,931,000	0	590,033	96,800	0	21,207
J1201	753,468	0	0	0	0	0	0	1,113,000	0	753,468	176,800	0	0
J1202	333,054	0	0	0	0	499,142	0	1,670,000	0	832,196	272,000	0	0
J1204	1,844,919	0	0	0	8,927,000	0	3,860,000	1,079,000	0	5,704,919	797,200	0	343,784
J1208	223,136	0	0	0	0	327,538	357,000	414,500	0	907,674	320,000	205,139	0
J1209	223,136	0	0	0	0	327,538	357,000	414,500	0	907,674	320,000	205,139	0
J1213	250,503	0	0	0	0	481,120	0	52,000	0	731,623	240,000	0	0
J1216	678,921	0	0	0	614,904	908,956	888,000	1,167,000	0	2,475,877	1,000,000	1,708,738	0
J1231	1,519,006	0	0	0	7,295,000	0	2,981,500	4,859,877	0	4,500,506	500,000	101,908	298,193
J1232	2,552,327	0	0	0	0	0	9,629,000	1,114,000	0	12,181,327	396,000	416,289	1,623,977
J1233	710,078	0	0	0	0	0	2,000,000	0	0	2,710,078	200,000	178,142	163,874
J1234	28,020,395	0	0	0	0	0	5,165,500	425,000	0	33,185,895	400,000	2,873,721	3,363,458
J1235	14,132,381	0	0	0	0	0	5,165,500	425,000	0	19,297,881	200,000	1,797,541	1,862,035

J1241	568,528	0	0	0	169,968	650,374	15,496,000	1,500,000	0	16,714,902	800,000	338,253	2,204,728
J1263	2,699,907	0	0	0	23,224,900	0	9,515,000	1,146,000	0	12,214,907	1,634,400	267,730	540,851
J1265	104,011	0	0	0	0	0	3,000,000	0	0	3,104,011	172,000	130,133	318,670
J1266	741,687	0	0	0	0	0	9,240,000	1,114,000	0	9,981,687	396,000	483,592	1,116,746
J1268	3,470,428	0	0	0	2,155,264	14,668,396	9,699,000	953,000	0	27,837,824	600,000	2,028,476	2,939,089
J1269	0	0	0	0	0	3,094,181	4,500,000	0	0	7,594,181	400,000	1,034,679	84,157
J1272	0	0	0	0	0	2,857,901	2,000,000	0	0	4,857,901	200,000	763,959	7,621
J1276	0	0	0	0	0	4,696,946	750,000	0	0	5,446,946	400,000	1,055,971	0
J1289	1,098,065	0	0	0	134,008	0	11,473,000	1,319,000	0	12,571,065	800,000	533,344	1,180,869
J1295	0	0	0	0	0	0	17,000,000	0	0	17,000,000	1,120,000	3,584,079	0
J1299	0	0	0	0	767,134	816,493	538,000	953,000	0	1,354,493	800,000	1,823,429	0
J1302	396,487	0	0	0	0	0	0	52,000	0	396,487	600,000	266,805	0
J1303	339,815	0	0	0	0	529,702	9,389,000	457,000	0	10,258,517	600,000	5,493,643	0
J1306	740,139	0	0	0	1,952,864	1,164,404	11,665,000	1,319,000	0	13,569,543	800,000	356,192	1,557,717
J1308	0	0	0	0	5,081,000	2,861,122	16,500,000	0	0	19,361,122	1,200,000	1,911,979	760,246
J1311	717,494	0	0	0	0	1,054,703	8,474,000	953,000	0	10,246,197	600,000	247,560	1,201,680
J1323	59,834	0	0	0	0	0	1,500,000	0	0	1,559,834	200,000	51,227	60,740
J1332	554,476	0	0	0	4,556,398	0	4,400,000	0	0	4,954,476	800,000	245,072	0
J1333	1,479,414	0	0	0	3,551,439	0	6,250,000	0	0	7,729,414	800,000	166,814	579,068
J1334	1,479,414	0	0	0	3,551,439	0	6,250,000	0	0	7,729,414	800,000	166,814	579,068
J1335	1,479,414	0	0	0	3,551,439	0	6,250,000	0	0	7,729,414	800,000	166,814	579,068
J1336	1,479,414	0	0	0	3,551,439	0	6,250,000	0	0	7,729,414	800,000	166,814	579,068
J1339	1,470,714	0	0	0	4,538,617	0	2,200,000	0	0	3,670,714	800,000	25,299	0
J1340	1,470,714	0	0	0	4,538,617	0	2,200,000	0	0	3,670,714	800,000	25,299	0
J1342	7,014,871	0	0	0	0	3,979,909	1,800,000	0	0	12,794,780	396,000	650,293	1,512,663
J1348	662,433	0	0	0	0	15,394,711	28,722,235	2,398,833	0	44,779,379	800,000	1,926,146	6,229,730
J1352	2,998,873	0	0	0	2,065,403	2,071,793	12,074,000	1,319,000	0	17,144,666	400,000	1,089,873	1,939,060
J1353	8,024,184	0	0	0	3,683,736	0	1,600,000	0	0	9,624,184	600,000	1,084,301	240,536
J1354	0	0	0	0	0	0	0	2,737,393	0	0	584,000	1,907,810	0
J1355	325,395	0	0	0	1,551,347	0	1,800,000	0	0	2,125,395	644,000	1,042,017	0
J1356	26,028,251	0	0	0	0	0	17,008,956	1,640,099	0	43,037,207	640,000	13,482,617	0

J1357	0	0	0	0	0	0	0	5,680,572	0	0	600,000	806,715	0
J1358	1,253,952	0	0	0	0	0	6,400,000	0	0	7,653,952	500,000	599,167	431,624
J1360	223,633	0	0	0	0	11,972,806	4,122,000	1,592,000	0	16,318,439	1,200,000	0	2,063,688
J1369	0	0	0	0	0	41,369,848	12,000,000	0	0	53,369,848	400,000	3,002,602	7,271,367
J1378	619,343	0	0	0	0	18,125,968	2,360,732	3,632,201	0	21,106,043	800,000	3,190,848	230,361
J1381	508,807	0	0	0	767,914	9,243,110	3,042,142	3,624,544	0	12,794,059	800,000	430,729	1,328,083
J1382	1,513,127	0	0	0	0	0	7,883,000	953,000	0	9,396,127	600,000	266,860	1,012,365
J1383	0	0	0	0	112,100	0	0	1,320,000	0	0	600,000	0	0
J1386	5,360,433	0	0	0	1,971,824	0	1,600,000	0	0	6,960,433	400,000	609,134	382,952
J1387	496,016	0	0	0	0	0	1,000,000	0	0	1,496,016	240,000	48,784	10,420
J1388	13,423,943	0	0	0	0	0	19,344,438	1,575,958	0	32,768,381	600,000	2,152,032	3,801,644
J1390	361,915	0	0	0	0	0	24,092,220	2,054,722	0	24,454,135	600,000	907,652	3,383,175
J1391	0	0	0	0	0	0	2,000,000	0	0	2,000,000	400,000	0	0
J1392	239,338	0	0	0	2,025,000	0	1,500,000	0	0	1,739,338	800,000	0	0
J1393	2,839,922	0	0	0	0	0	2,000,000	0	0	4,839,922	800,000	113,462	54,522
J1396	0	0	0	0	0	0	0	0	0	0	600,000	5,372,281	0
J1405	43,152,476	0	0	0	0	0	18,686,577	1,728,951	0	61,839,053	560,000	5,077,413	6,730,398
J1407	7,306,100	0	0	0	0	4,020,091	1,800,000	0	0	13,126,191	400,000	667,127	1,558,111
J1408	1,657,124	0	0	0	11,607,321	0	7,509,000	1,166,000	0	9,166,124	400,000	1,942,455	0
J1419	622,507	0	0	0	0	368,090,392	25,068,952	2,376,304	0	393,781,851	1,000,000	10,824,161	66,932,209
J1422	176,733	0	0	0	0	0	9,798,000	292,368	0	9,974,733	600,000	568,435	826,512
J1447	2,935,597	0	0	0	6,578,192	0	25,000,000	0	0	27,935,597	1,000,000	2,212,153	2,374,966
J1448	1,822,858	0	0	0	8,026,000	0	2,981,500	4,859,877	0	4,804,358	600,000	135,733	225,138
J1450	1,858,136	0	0	0	7,579,000	0	2,165,000	4,859,877	0	4,023,136	600,000	167,195	37,432
J1453	9,501,934	0	0	0	0	0	6,275,000	652,000	0	15,776,934	753,200	3,137,663	0
J1454	374,076	0	0	0	0	0	0	52,000	0	374,076	234,000	667,027	0
J1464	5,998,010	0	0	0	230,426,930	0	10,806,000	1,319,000	0	16,804,010	2,371,200	0	989,602
J1466	0	0	0	0	0	0	0	2,112,044	0	0	1,000,000	3,622,075	0
J1470	0	0	0	0	0	3,522,322	0	0	0	3,522,322	119,200	55,764	529,500
J1475	287,053	0	0	0	93,000	0	0	52,000	0	287,053	300,000	0	0
J1481	508,807	0	0	0	767,914	9,243,110	5,082,564	3,452,480	0	14,834,481	800,000	430,729	1,736,168



J1482	418,607	0	0	0	0	1,407,580	10,000,000	0	0	11,826,187	600,000	900,269	864,968
J1488	36,798,935	250,000	0	0	12,755,599	8,894,449	995,500	659,500	0	46,938,884	2,000,000	1,872,099	5,515,678
J1490	73,597,869	250,000	0	0	25,511,202	17,788,899	995,500	659,500	0	92,632,268	4,000,000	3,694,198	10,832,256

*Values in RED, under the TO Network Upgrades column, denote estimates from Phase I (As Phase 2 ICIF Studies were not complete prior to this posting).



Analyses performed demonstrate the following transmission facilities are required to reliably interconnect this group of generators to the transmission system. Energy Resource Interconnection Service (ERIS) Network Upgrades and Network Resource Interconnection Service (NRIS) Network Upgrades are shown in Table 3. Shared Network Upgrades are shown in Table 4.

Table 3: ERIS & NRIS Upgrades (Planning level cost estimates)

Network Upgrade	Operating Company (TO)	Projects in Violation	Planning Level Cost Estimate (\$)	Network Upgrade Type	TO Self-Fund
J1039 - Montgomery 345 kV Bus Upgrade	AMMO	J1437 J1488 J1490	250,000.00	ERIS Thermal	Yes
Tower Road Breaker / Terminal Upgrades	NIPSCO	J1333 J1334 J1335 J1336 J1339 J1340	1,400,000.00	ERIS Thermal	No
J1232 - Arland 138kV Line Rebuild	AMMO	J1198 J1204 J1232 J1263 J1382	3,200,000.00	ERIS Thermal	Yes
Belle Tap-Gasco Tap 138 kV Line Rebuild	AMMO	J1488 J1490	4,500,000.00	ERIS Thermal	Yes
Belle Tap-Meta Tap 138 kV Line Rebuild	AMMO	J1488 J1490	14,000,000.00	ERIS Thermal	Yes
Miller-Meta Tap 138 kV Line Rebuild	AMMO	J1488 J1490	7,200,000.00	ERIS Thermal	Yes
Bland-Gasco Tap 138 kV Line Rebuild	AMMO	J1488 J1490	5,200,000.00	ERIS Thermal	Yes
McCredie - Overton 345 kV Line Rebuild	AECI/AMMO	J1198 J1201 J1202 J1204 J1208 J1209 J1213 J1216 J1232 J1241 J1266 J1268 J1289 J1302 J1303 J1306 J1311 J1352 J1382 J1408 J1422 J1464 J1488 J1490	36,000,000.00	ERIS Thermal	Yes
Pike - New Pike 69 kV Reconductor	AMMO	J1268	35,000.00	ERIS Thermal	Yes
Warrenton 161 kV Bus Tie 1-2 Upgrade	AMMO	J1488 J1490	250,000.00	ERIS Thermal	Yes
Overton - Sedalia East 161 kV line Upgrade	AMMO	J1191 J1488 J1490	7,000,000.00	ERIS Thermal	Yes
Warrenton - Montgomery-3 161 kV line Rebuild	AMMO	J1488 J1490	13,000,000.00	ERIS Thermal	Yes
McBaine Tap - Overton 161 kV line Rebuild	AMMO	J1191 J1268 J1352 J1488 J1490	8,500,000.00	ERIS Thermal	Yes
Montgomery - Guthrie-3 161 kV line Rebuild	AMMO	J1352 J1488 J1490	27,000,000.00	ERIS Thermal	Yes

New Moro - Cottage Hills 138 kV Line + Cottage Hills - Laclede 138kV Line Upgrade	AMIL	J1198 J1200 J1201 J1204 J1232 J1234 J1235 J1263 J1265 J1266 J1289 J1323 J1332 J1333 J1334 J1335 J1336 J1339 J1340 J1342 J1348 J1353 J1355 J1356 J1358 J1360 J1378 J1381 J1382 J1386 J1387 J1390 J1392 J1405 J1407 J1408 J1419 J1447 J1453 J1454 J1464 J1475 J1481 J1482	23,000,000.00	ERIS Thermal	Yes
Latham - Docket 138 kV line Rebuild	AMIL	J1289 J1408 J1453 J1454 J1488 J1490	4,000,000.00	ERIS Thermal	Yes
Troy - J1268 161 kV line Rebuild	AMIL	J1268	2,000,000.00	ERIS Thermal	Yes
J1453 - Havana 138 kV line Rebuild	AMIL	J1453 J1454	8,500,000.00	ERIS Thermal	Yes
J1450 - ReidHV 161kV Line Upgrade	BREC	J1231 J1448 J1450	5,200,000.00	ERIS Thermal	No
Oakland City - 1388 POI 138kV Line Rebuild	DEI	J1388	13,423,943.00	ERIS Thermal	No
Washington Municipal - J1234 POI 138kV Line Rebuild	DEI	J1234 J1235 J1405	13,674,815.00	ERIS Thermal	No
Edwards - Peabody Tap 138kV Line Rebuild	DEI	J1234 J1235 J1405	20,000.00	ERIS Thermal	No
Edwards - Cranes 138kV Line Rebuild	DEI	J1234 J1235 J1356 J1405	35,280,000.00	ERIS Thermal	No

Vincennes Jct. - Washington Municipal 138kV Line Rebuild	DEI	J1234 J1235 J1405	12,900,000.00	ERIS Thermal	No
Cranes - Bedford 138kV Line Rebuild	DEI	J1234 J1235 J1356 J1405	48,316,000.00	ERIS Thermal	No
Loop Cato Tap - Duff 138kV Line In & Out of Cato 138kV Substation.	SIGE	J1393 J1233	3,550,000.00	ERIS Thermal	No
Loop Cato Tap - Duff 138kV Line In & Out of Cato 138kV Substation.	SIGE	J1393 J1233		ERIS Thermal	No
Loop Cato Tap - Duff 138kV Line In & Out of Cato 138kV Substation.	SIGE	J1393 J1233		ERIS Thermal	No
Loop Cato Tap - Duff 138kV Line In & Out of Cato 138kV Substation.	SIGE	J1393 J1233		ERIS Thermal	No
Loop Cato Tap - Duff 138kV Line In & Out of Cato 138kV Substation.	SIGE	J1393 J1233		ERIS Thermal	No
Loop Cato Tap - Duff 138kV Line In & Out of Cato 138kV Substation.	SIGE	J1393 J1233		ERIS Thermal	No
Loop Cato Tap - Duff 138kV Line In & Out of Cato 138kV Substation.	SIGE	J1393 J1233		ERIS Thermal	No
Loop Cato Tap - Duff 138kV Line In & Out of Cato 138kV Substation.	SIGE	J1393 J1233		ERIS Thermal	No
Aetna - Praxair3 138kV Line Upgrade	NIPS	J1333 J1334 J1335 J1336 J1339 J1340 J1342 J1353 J1358 J1386 J1387 J1407 J1447	8,000,000.00	ERIS Thermal	No
Aetna - West Mill 138kV Line Upgrade	NIPS	J1342 J1353 J1358 J1386 J1387 J1407	1,000,000.00	ERIS Thermal	No
Ainsworth - Green Acres 138kV Line Upgrade	NIPS	J1353 J1386 J1447	12,120,000.00	ERIS Thermal	No
Flint Lake - South Valparaiso 138kV Line Upgrade	NIPS	J1342 J1407	12,500,000.00	ERIS Thermal	No

Green Acres - St John 138kV Line Upgrade (Remove Sag Limit)	NIPS	J1332 J1333 J1334 J1335 J1336 J1339 J1340 J1353 J1386 J1387 J1447	1,000,000.00	ERIS Thermal	No
Henderson 161/138kV Xfmr #1 + Breaker/Switch/Linetrap Upgrades	BREC	J1269 J1272 J1276 J1308	5,200,000.00	NRIS Thermal	No
Schahfer tap - Schahfer 138kV Line Rebuild/Upgrade	NIPS	J1342 J1407	8,000,000.00	NRIS Thermal	No
Vandalia - Pike 69 kV Line Upgrade	AMMO	J1268	14,000,000.00	NRIS Thermal	Yes
New London - Kokomo (Highland Park) 230kV #2 New Line (Parallel Line Addition)	DEI	J1378 J1381 J1481 J1482	24,000,000.00	NRIS Thermal	No
Culley - Dubois 138kV Path Rebuild/Upgrade	SIGE	J1269 J1272 J1276 J1369	49,680,000.00	NRIS Thermal	No
Hortonville - Edwardsport 345kV New Line Addition	DEI	J1348 J1360 J1378 J1419	413,000,000.00	NRIS Thermal	No
Spencer Creek – Maywood 345kV Line Rebuild	AMMO	J1202 J1208 J1209 J1213 J1216 J1241 J1268 J1299 J1303 J1306 J1311 J1352 J1437 J1488 J1490	37,000,000	NRIS Thermal	Yes
25 Mvar 161 kV capacitor bank at Overton 161 kV	AMIL	J1488 J1490	500,000.00	ERIS Voltage (Ameren LPC)	Yes

Table 4: Shared Network Upgrades (Planning level cost estimates)

Shared Network Upgrade	TO	Higher queued projects associated with SNU	Study projects associated with SNU	Cost of solution (\$)
No Projects Met Criteria – N/A				

Note:

- 1) Details pertaining to upgrades, costs, and the execution plan for interconnection of the generating facility at the POI will be documented in the Facility Study for Interconnecting Generator.
- 2) Facilities that have been included as base case assumptions and the level of interconnection service that would be conditional upon these facilities being in service will be documented in the GIA (Generator Interconnection Agreement) for each respective GI request successfully

2. FERC Order 827 Compliance Review

The Final Rule of FERC Order 827 “Reactive Power Requirements for Non-Synchronous Generation”, which was issued June 16, 2016, stated that “Under this Final Rule, newly interconnecting non-synchronous generators that have not yet executed a Facilities Study Agreement as of the effective date of this Final Rule will be required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.” As such, this Final Rule applies to all non-synchronous (wind, solar, and battery storage) projects included in the DPP 2018 April Central study cycle.

In this study, the power factor at the high-side of the generator substation for each inverter based project was calculated and reviewed. The study method is to set Qgen of each study project at its Qmax, solve the case, then record the P and Q injection on the high side of the generator substation to calculate the lagging power factor (injecting VAR to the system). The same process is then repeated by setting Qgen at Qmin to calculate the leading power factor (absorbing VAR from the system).

The results show that all projects meet the requirement to maintain 0.95 leading power factor, however, four projects do not meet the requirement to provide reactive power capability corresponding to 0.95 lagging power factor, as highlighted in red below in **Table 5**. Additional reactive support will be needed for these projects to meet the FERC requirement on reactive power capability prior to the completion of their GIA.

Table 5: FERC Order 827 Review Results

FERC Order 827				Steady State (At Generator Substation)							
Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Requirement?	Add'l VAR Needed (MVar)
				P (MW)	Q (MVar)	p.f. (pu)	P (MW)	Q (MVar)	p.f. (pu)		
J1191	66.0	±11.45	None	65.3	23.1	0.9428	65.1	-26.6	-0.9257	Yes	
J1198	GEN: 51.8 GEN1: 13.1	GEN: ±17.0260 GEN1: ±4.3060	None	64.9	21.1	0.9510	64.9	-21.6	-0.9488	No	0.2
J1199	GEN: 20.0 GEN1: 4.8	GEN: ±6.5740 GEN1: ±1.5780	None	24.8	7.9	0.9528	24.7	-7.8	-0.9536	No	0.3
J1200	GEN: 24.2 GEN1: 6.2	GEN: ±7.9540 GEN1: ±2.0380	None	9.4	10.2	0.6777	30.1	-10.7	-0.9422	Yes	
J1201	GEN: 44.2 GEN1: 11.1	GEN: ±14.5280 GEN1: ±3.6480	None	54.9	19.1	0.9445	54.9	-17.3	-0.9538	Yes	
J1202	GEN: 68.4 GEN1: 16.6	GEN: ±22.4820 GEN1: ±5.4560	None	83.6	28.8	0.9455	83.6	27.1	-0.9513	Yes	
J1204	202.9	±30.346	1 x 75 MVar Cap	199.6	73.1	0.9390	199.1	-74.1	-0.9372	Yes	
J1208	82.0	±27.027	1 x 13 MVar Cap	81.1	26.4	0.9509	80.8	-48.0	-0.8597	No	0.3
J1209	82.0	±27.027	1 x 13 MVar Cap	81.1	25.9	0.9526	80.8	-48.7	-0.8565	No	0.8
J1216	GEN: 92.5 GEN1: 92.5	GEN: ±41.0860 GEN1: ±41.0860	GEN: 1 x 4 MVar Cap GEN1: 1 x 4 MVar Cap	183.8	70.7	0.9333	183.3	-114.8	-0.8475	Yes	
J1231	125.0	±41.0860	None	123.8	24.8	0.9805	123.4	-64.5	-0.8862	No	15.9
J1232	100.8	±37.5800	3 x 7 MVar Cap	99.4	43.6	0.9158	98.8	-64.6	-0.8370	Yes	
	GEN: 32.5	±16.4340	1 x 2 MVar Cap	160.4	93.0	0.8651	159.8	-89.8	-0.8718	Yes	

J1233 & J1393	GEN1: 130	±65.7370	1 x 4 MVar Cap								
J1234	102.4	±43.46	1 x 20 MVar Cap	101.3	45.7	0.9115	100.5	-76.8	-0.7946	Yes	
J1235	50.99	±20.38	1 x 10 MVar Cap	49.9	21.3	0.9197	49.3	-38.3	-0.7897	Yes	
J1241	171.3	±57.1	None	169.7	40.5	0.9727	169.4	-80.3	-0.9036	No	15.3
J1263	GEN: 135.0 GEN1: 136.8 GEN2: 136.8	GEN: ±44.3720 GEN1: ±44.9640 GEN2: ±44.9640	GEN: 1 x 20 MVar Cap GEN1: 1 x 20 MVar Cap GEN2: 1 x 20 MVar Cap	401.9	128.5	0.9525	399.6	-239.9	-0.8574	No	3.6
J1265	43.7	±17.4080	1 x 7.2 MVar Cap	43.1	18.1	0.9220	42.8	-30.1	-0.8180	Yes	
J1266	100.8	±37.5800	3 x 5.1 MVar Cap	99.3	38.2	0.9333	98.6	-64.5	-0.8368	Yes	
J1268	165.0	±64.6700	None	162.9	55.4	0.9467	161.7	-116.7	-0.8109	Yes	
J1269	GEN: 123.56 GEN1: 20.59	GEN: ±16.45 GEN1: ±16.45	None	142.5	26.5	0.9831	142.0	-71.6	-0.8929	No	20.3
J1272	53.5	±40.125	None	52.9	28.8	0.8783	52.4	-61.6	-0.6479	Yes	
J1276	106.9	GEN: ±57.0	None	106.1	51.7	0.8990	105.6	-66.0	-0.8480	Yes	
J1289	200.0	±96.864	1 x 15 MVar Cap	195.0	84.3	0.9179	192.6	-148.2	-0.7925	Yes	
J1295	280.0	±158.683	None	275.3	98.7	0.9413	271.8	-263.4	-0.7181	Yes	
J1299	149.0	±72.0	None	146.8	52.1	0.9424	146.1	-103.4	-0.8163	Yes	
J1302	150.0	±85.009	None	147.4	54.5	0.9379	144.9	-151.6	-0.6910	Yes	
J1303	97.0	±47.0	None	95.7	34.9	0.9395	95.4	-63.4	-0.8329	Yes	
J1306	200.0	±113.345	None	196.6	73.1	0.9373	194.5	-184.0	-0.7264	Yes	
J1308	300.0	±170.018	None	294.9	108.7	0.9383	290.7	-285.1	-0.7140	Yes	
J1311	GEN: 78.98 GEN1: 78.98	GEN: ±25.000 GEN1: ±25.000	GEN: 4 x 5 MVar Cap GEN1: 4 x 5 MVar Cap	156.4	98.7	0.8457	156.1	-58.4	-0.9366	Yes	
J1323 & J1392	GE1N: 25	GEN1: ±16.4341	GEN1: 1 x 2 MVar Cap	123.6	101.7	0.7722	123.3	-79.7	-0.8398	Yes	
	GEN: 100	GEN: ±65.737	GEN: 1 x 4 MVar Cap								
J1332	200.0	±82.155	None	196.8	44.1	0.9758	195.1	-146.4	-0.7999	No	20.6
J1333, J1334, J1335, & J1336	GEN: 249.5 GEN1: 44.04	GEN: ±54.36 GEN1: ±11.64	None	813.5	172.3	0.9783	809.5	-388.3	-0.9016	No	95.1
	GEN2: 249.5 GEN3: 44.04	GEN2: ±54.36 GEN3: ±11.64	None								
	GEN4: 249.5 GEN5: 44.04	GEN4: ±54.36 GEN5: ±11.64	None								
	GEN6: 249.5 GEN7: 44.04	GEN6: ±54.36 GEN7: ±11.64	None								

J1339 & J1340	GEN: 249.5 GEN1: 44.04	GEN: ±56.1 GEN1: ±9.9	None	402.8	70.7	0.9849	400.1	-215.6	-0.8803	No	61.7
	GEN: 249.5 GEN1: 44.04	GEN: ±56.1 GEN1: ±9.9	None								
J1342	99.0	±59.529	None	98.0	45.8	0.9059	97.4	-80.5	-0.7708	Yes	
J1348	200.0	±82.155	None	196.9	46.6	0.9731	195.4	-141.4	-0.8101	No	18.1
J1352	100.0	±41.078	None	98.2	29.2	0.9585	97.7	-57.2	-0.8630	No	3.1
J1353	151.8	±91.2780	None	150.3	69.5	0.9077	149.5	-123.4	-0.7712	Yes	
J1354	118.0	±88.6320	None	117.2	-25.4	0.9773	116.4	-110.8	-0.7243	No	63.9
J1355	131.0	±97.8920	None	129.8	81.6	0.8466	129.3	-122.0	-0.7273	Yes	
J1356	135.0	±96.5690	None	133.7	81.1	0.8550	132.2	-132.1	-0.7074	Yes	
J1357	151.8	±91.2780	None	150.3	70.7	0.9049	149.4	-126.0	-0.7644	Yes	
J1358	125.0	±75.1630	None	123.7	57.9	0.9057	123.1	-101.2	-0.7725	Yes	
J1360	GEN: 150.87 GEN1: 150.87	GEN: ±73.0700 GEN1: ±73.0700	GEN: 1 x 15 MVar Cap GEN1: 1 x 15 MVar Cap	292.7	147.5	0.8930	287.8	-216.3	-0.7994	Yes	
J1369	101.4	±42.8770	None	100.1	24.1	0.9722	99.5	-73.2	-0.8055	No	8.8
J1378	200.0	±94.0140	None	196.7	63.2	0.9521	195.0	-141.2	-0.8100	No	1.5
J1381	200.0	±94.0140	None	196.6	62.3	0.9533	195.4	-138.1	-0.8166	No	2.3
J1382	150.9	±87.144	None	148.5	70.7	0.9029	146.4	-128.1	-0.7526	Yes	
J1383	150.9	±87.144	None	148.3	68.2	0.9085	146.9	-122.8	-0.7672	Yes	
J1386	101.4	±42.8770	None	100.2	33.9	0.9473	99.6	-72.8	-0.8073	Yes	
J1387	61.5	±20.4000	None	60.8	26.7	0.9156	60.7	-15.7	-0.9681	Yes	
J1388	155.2	±51.0100	None	153.2	26.1	0.9858	152.3	-93.3	-0.8527	No	24.3
J1390	150.9	±87.144	None	148.3	67.6	0.9099	146.7	-125.5	-0.7599	Yes	
J1391	101.6	±58.6890	None	99.8	45.2	0.9109	98.8	-84.1	-0.7615	Yes	
J1396	61.0	±29.0	None	60.6	24.4	0.9276	60.5	35.1	-0.8650	Yes	
J1405	141.8	±56.5040	None	140.3	43.9	0.9544	138.9	-104.9	-0.7980	No	2.2
J1407	101.6	±58.6890	None	99.8	44.5	0.9133	98.5	-87.5	-0.7476	Yes	
J1408	GEN: 100.8 GEN1: 100.98 GEN2: 102.91	GEN: +50.526 / -31.302 GEN1: +17.261 / -17.261 GEN2: +17.261 / -27.0	None	99.6	74.0	0.8027	99.2	-100.2	-0.7036	Yes	
J1419	253.2	±103.212	None	250.1	73.7	0.9592	248.8	-169.7	-0.8261	No	8.5
J1422	GEN: 2 x 76.0 GEN1: 2 x 76.0	GEN: 2 x ±25.0 GEN1: 2 x ±25.0	GEN: 1 x 25 MVar Cap GEN1: 1 x 25 MVar Cap	149.8	139.2	0.7325	146.3	-177.9	-0.6352	Yes	
J1447	250.0	±82.171	2 x 17 MVar Cap	247.0	75.5	0.9563	245.9	-140.7	-0.8680	No	5.7
J1448	150.0	±49.3030	2 x 11 MVar Cap	148.5	50.5	0.9468	147.8	-84.4	-0.8684	Yes	
J1450	150.0	±49.3030	2 x 11 MVar Cap	148.4	50.2	0.9473	147.8	-83.7	-0.8702	Yes	

J1453	165.0	±99.063	None	158.7	95.3	0.8573	152.6	-150.2	-0.7127	Yes	
J1454	40.9	±30.7760	None	40.2	27.8	0.8225	40.0	-35.6	-0.7470	Yes	
J1464	GEN: 150.375 GEN1: 150.375 GEN2: 150.375 GEN3: 150.375	GEN: ±49.4260 GEN1: ±49.4260 GEN2: ±49.4260 GEN3: ±49.4260	None	589.4	530.1	0.7435	590.1	75.6	-0.9919	Yes	
J1466	250.0	±82.171	2 x 17 MVar Cap	247.0	80.6	0.9507	245.7	-142.1	-0.8657	No	0.6
J1475	75.0	Max: +39.1320 Min: -20.3500	2 x 4 MVar Cap	75.0	39.1	0.8867	75.0	20.3	-0.9653	Yes	
J1481	200.0	±159.158	None	196.6	125.6	0.8427	194.2	-220.1	-0.6616	Yes	
J1482	150.0	Qmax: +74.6110 Qmin: -62.838	None	148.0	55.9	0.9355	147.4	-90.0	-0.8535	Yes	

*Items in **RED** text denote observed MVAR modeling differences between what was approved as part of the A10 Model Review process (In Phase I) and what's in each project's application. These values will need Interconnection Customer verification prior to Phase 3 System Impact Studies beginning. The DPP Phase 3 model build period will need to verify these values.

3. Model Development and Study Assumptions

3.1. Base Case Models

The origin of the DPP 2019 Study Cycle Central models is the MTEP 19 models with the Bench Cases including all pre-queued projects and associated network upgrades, while the Study Cases contain all of the interconnection requests in DPP 2018 April Central Phase II, in addition to all the facilities in the Bench Cases.

- Bench Cases

- DPP19-Central-Phase_II-2024SH-Bench_10292021_FINAL.raw
- DPP19-Central-Phase_II-2024SUM-Bench_10292021_FINAL.raw

- Study Cases

- DPP19-Central-Phase_II-2024SH-Study-Charging_10292021_FINAL.raw
- DPP19-Central-Phase_II-2024SH-Study-Discharging_10292021_FINAL.raw
- DPP19-Central-Phase_II-2024SUM-Study-Charging_NIPSCO_LPC_10292021_FINAL.raw
- DPP19-Central-Phase_II-2024SUM-Study-Discharging_10292021_FINAL.raw

3.2. Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

3.3. Contingencies

The following contingencies were considered in the steady state analysis:

- 1) NERC Category P0 (system intact -- no contingencies)
- 2) NERC Category P1 contingencies

- a. Single element outages, at buses with a nominal voltage of 68 kV and above
 - b. Multiple element NERC Category P1 contingencies
- 3) NERC Category P2-P7 contingencies
- 4) For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

3.4. Study Methodology

Non-linear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP 2019 Central generating facilities was evaluated by comparing the steady state performance of the transmission system in the Bench and Study Cases. Analyses used PSS®E version 33.7.0 and TARA version 2001.

3.5. Performance Criteria

A branch is considered a thermal constraint if the following conditions are met:

- 1) The generator has a larger than twenty percent (20%) sensitivity factor on the overloaded facilities under post-contingent condition (see NERC TPL) or five percent (5%) sensitivity factor under system-intact condition, or
- 2) The overloaded facility or the overload-causing contingency is at generator's outlet, or
- 3) The megawatt impact due to the generator is greater than or equal to twenty percent (20%) of the applicable rating (normal or emergency) of the overloaded facility, or
- 4) For any other constrained facility, where none of the Study Generators meet one of the above criteria, however, the cumulative MW impact of the group of study generators is greater than twenty percent (20%) of the rating of the facility, then only those study generators whose individual MW impact is greater than five percent (5%) of the rating of the facility and has DF greater than five percent (5%) will be responsible for mitigating the cumulative MW impact constraint, or
- 5) Impacts on Affected Systems would be classified as Injection constraints based on the Affected Systems' criteria, or
- 6) Any other applicable Transmission Owner FERC filed Local Planning Criteria

are met. A bus is considered a voltage constraint if both of the following conditions are met:

- 1) The bus voltage is outside of the applicable normal or emergency limits for the post change case, and
- 2) The change in bus voltage is greater than 0.01 per unit

All generators must mitigate thermal injection constraints and voltage constraints in order to obtain any type of Interconnection Service. Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the Deliverability algorithm, to meet the system performance criteria for NERC category P1 events, if DFAX due to the study generator is equal to or greater than 5%.

4. Thermal Analysis

The MISO Thermal ERS & Ameren Thermal ERS LPC study analysis results for the 2019 Central Study Cycle show the following generator projects causing constraints:

J1191, J1198, J1200, J1201, J1202, J1204, J1208, J1232, J1208, J1209, J1213, J1216, J1231, J1232, J1233, J1234, J1235, J1241, J1263, J1265, J1266, J1268, J1289, J1302, J1303, J1306, J1311, J1323, J1332, J1333, J1334, J1335, J1336, J1339, J1340, J1342, J1348, J1352, J1353, J1355, J1356, J1358, J1360, J1378, J1381, J1382, J1386, J1387, J1388, J1390, J1392, J1393, J1405, J1407, J1408, J1419, J1422, J1447, J1448, J1450, J1453, J1454, J1464, J1475, J1481, J1482, J1488, J1490

The details pertaining to the thermal analysis can be found in **Appendix A – Ameren System Impact Study Report (CEII)**, **Appendix C – MISO ERS Analysis (CEII)**, and **Appendix F - Ameren LPC Results (CEII)**.

5. Voltage Analysis

The MISO Voltage ERS & Ameren Voltage ERS LPC study analysis results for the 2019 Central Study Cycle show the following generator projects causing violations:

J1488 & J1490

The details pertaining to the voltage analysis can be found in **Appendix A – Ameren System Impact Study Report (CEII)**, **Appendix F - Ameren LPC Results (CEII)**, **Appendix B – Cost Allocation Summary (CEII)**, and **Appendix C – MISO ERS Analysis (CEII)**.

6. Stability Analysis

The MISO DPP Stability analysis shows that the study projects did not adversely impact the system to where a network upgrade would be needed to mitigate observed stability related violations.

The details pertaining to the MISO stability analysis can be found in **Appendix H - MISO Stability Analysis (CEII)**.

Initially observed stability concerns:

- J1289 (failed to ride through several faults, causing a TSAT convergence issue)
- J1268/J1269 – Trips offline under various fault simulations
- Various Project Plot Concerns – J1213, J1335, J1408, J1447, J1450, J1453, J1454, J1466 (Oscillations/Damping/Gen Rise/Drop Concerns).

After some offline discussions, the critical concerns above have been fixed through model tuning. Updated models have been provided to MISO for use in Phase 3 studies.

An additional stability study capturing the Ameren Local Planning Criteria (LPC) for new generation interconnections was also performed by Ameren and also shows that study projects did not adversely impact the system under the Ameren LPC for new generation interconnections. The details pertaining to the stability analysis can be found in **Appendix G - Ameren Stability Analysis (CEII)**.

Ameren LPC Stability Conclusions:

Based on the simulations performed in this study, the performance of the MISO projects J1213, J1216, J1232, J1241, J1266, J1311, J1382, J1383, J1422, J1475, J1464, J1488 and J1490 were found to be acceptable under the fault scenarios prescribed by the Ameren Planning Criteria and Guidelines. Performance issues were identified for the other projects evaluated as discussed below. No small signal damping issues or transient voltage recovery violations were identified in this study.

The dynamic models provided for projects J1289, J1360, J1453 and J1454 produced oscillations in the base case during a no-fault simulation. The WTDTAU1 renewable generator drive train models associated with projects J1289 and J1360 had to be removed. The TP parameter (electrical power filter time constant) had to be increased to 0.05 in the WTTQAU1 torque controller models associated with projects J1453 and J1454 to prevent electrical power oscillations in these two generator projects.

Ameren was not able to evaluate the voltage and frequency ride-through capability of MISO project J1454 because the generator interconnection customer did not provide data to model voltage and frequency relays.

MISO projects J1199, J1200, J1201, J1289 and J1464 were unable to meet the voltage schedule required by Ameren. Additional reactive compensation is needed to meet these requirements.

The frequency protection relays associated with MISO projects J1199, J1200, J1201, J1202, J1204, J1268, J1289, J1299, J1302 and J1306 included instantaneous or very short durations for the trip settings. These settings did not allow these generators to remain on-line for 9 cycles as required by FERC Order 661A. In addition, NERC strongly discourages the use of instantaneous trip settings in frequency relays due to the potential for false tripping. Therefore, these settings must be modified to have a duration of at least 9 cycles.

The voltage protection relays associated with MISO projects J1198, J1200, J1201, J1208, J1209, J1263, J1268, and J1352 tripped these generators due to low or high voltages during the fault simulations. Additional dynamic reactive capability is recommended for projects J1200, J1263 and J1352. The other trips can be avoided by increasing the duration for the trip settings to at least 9 cycles or by reducing the breaker failure clearing times in the interconnection substation.



Sustained oscillations were observed in the real and reactive power output for the J1408 study generators during the fault simulations. These oscillations are unacceptable and must be resolved by the generator owner. Design changes in the generation facility may be required to eliminate these oscillations. This is not an issue that the transmission owner can resolve.

Remote trips at several wind farms connected to Ameren or nearby transmission systems were observed. Specially the White Oak, Bishop Hill, California Ridge and W2-048 wind farms tripped for the faults evaluated in this study. These trips are not attributable to the study generation but to the specific settings of voltage and frequency protection relays associated with these generation facilities. It is recommended that MISO follow up with these generation owners to resolve these potential tripping occurrences.

The model deficiencies, oscillations and the unacceptable relay settings identified in this study must be corrected. Acceptable models must be provided before Phase III of the system impact study begins. If these issues are not corrected, then the generator performance will be considered unacceptable.

7. Short Circuit Analysis

The short circuit analysis results for 2019 Central study cycle shows that the study generators do not cause any short circuit violations. The details pertaining to the short circuit analysis can be found in **Appendix I – Short Circuit Studies (CEII)**.

8. Affected System Impact Study

The details pertaining to the AECl, PJM, SPP, and TVA Affected Systems studies are in **Appendix J – Affected System Studies (CEII)**.

Please refer to the appendix above for all details associated with the constraints, mitigations, and costs assigned to each project listed in the sections below. You can also refer to **Appendix B - Cost Allocation Summary (CEII).xlsx** (AFS Cost Sharing tab) for an overall summary of all AFS costs for each study project associated with this study cycle.

Projects flagged on Affected Systems Constraints:

SPP AFS Report

J1299

TVA AFS Report

J1231, J1448, J1450

AECl AFS Report

J1191, J1216, J1241, J1268, J1289, J1299, J1306, J1315, J1352, J1437, J1456, J1464, J1471, J1488, J1490

PJM AFS Report

J1197, J1200, J1204, J1226, J1263, J1289, J1308, J1332, J1333, J1334, J1335, J1336, J1339, J1340, J1353, J1355, J1381, J1383, J1386, J1392, J1408, J1447, J1464, J1475, J1481

9. Deliverability Analysis

9.1. Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted NRIS. If the generator is deemed not fully deliverable, the customer can choose either to change the project to an Energy Resource (ER) project or to proceed with the system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up. The wind generators are tested at 100% of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

MISO Generator Deliverability Study whitepaper describing the algorithm can be found in BPM 015 – Generation Interconnection, Appendix C.

9.2. Determining the MW Restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study has a DF greater than 5%, part or all of its output is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator’s output is not deliverable.

9.3. Deliverability Study Results

The limiting constraints (mon-con pairs) seen in the deliverability analysis for the 2019 Central Study Cycle are summarized in Appendix D - MISO Deliverability Analysis (CEII). This appendix also includes a deliverability breakdown for each project in the study cycle (where the deliverability level, constraint list, DFAX, level of service attainable, and assigned costs are listed).

Please refer to the above appendix for all detailed MISO NRIS Deliverability information. The table below summarizes each project’s current deliverability level and the number of constraints it’s been flagged on. Overall costs are captured in **Table 2** of this report and are also included within each project tab in **Appendix D**.

Table 6: Project Deliverability Summary

Project #	# of Constraints	Deliverability Level (MW)	% Deliverable
J1191	0	64	100
J1198	0	51.8	100
J1199	0	20	100
J1200	0	24.2	100
J1201	0	44.2	100
J1202	1	66.86	97.75
J1204	0	199.3	100
J1208	1	78.19	97.74
J1209	1	78.19	97.74
J1213	1	58.64	97.73
J1216	1	180.82	97.74

J1231	0	125	100
J1232	0	99	100
J1233	0	32.5	100
J1234	0	100	100
J1235	0	50	100
J1241	1	161.27	97.74
J1263	0	408.6	100
J1265	0	43	100
J1266	0	99	100
J1268	2	78.73	89.47
J1269	2	0	0
J1272	2	0	0
J1276	2	0	0
J1289	0	200	100
J1295	0	280	100
J1299	1	145.63	97.74
J1302	0	150	100
J1303	1	92.85	97.74
J1306	1	195.48	97.74
J1308	1	0	0
J1311	1	146.6	97.73
J1323	0	25	100
J1332	0	200	100
J1333	0	199.5	100
J1334	0	199.5	100
J1335	0	199.5	100
J1336	0	199.5	100
J1339	0	199.5	100
J1340	0	199.5	100
J1342	1	91.12	92.04
J1348	1	120.50	60.25
J1352	1	97.74	97.74
J1353	0	150	100
J1354	0	118	100
J1355	0	131	100
J1356	0	74	100
J1357	0	150	100
J1358	0	125	100
J1360	1	180.74	60.25
J1369	1	89.57	89.57
J1378	2	0	0
J1381	1	154.86	77.43
J1382	0	150	100

J1383	0	150	100
J1386	0	100	100
J1387	0	No NRIS Election	N/A
J1388	0	No NRIS Election	N/A
J1390	0	150	100
J1391	0	100	100
J1392	0	100	100
J1393	0	130	100
J1396	0	60	100
J1405	0	140	100
J1407	1	92.04	92.04
J1408	0	100	100
J1419	1	68.41	27.36
J1422	0	150	100
J1447	0	250	100
J1448	0	150	100
J1450	0	150	100
J1453	0	165	100
J1454	0	40.9	100
J1464	0	592.8	100
J1466	0	250	100
J1470	1	18.07	60.23
J1475	0	75	100
J1481	1	154.86	77.43
J1482	1	0	0
J1488	1	488.7	97.74
J1490	1	977.39	97.74

NRIS/Deliverability Considerations:

- If a project owner drops their NRIS requested level to 0MW, any NRIS/Deliverability assigned costs (Due to the project being flagged on certain network upgrades) would be removed. This move would guarantee a project would not be assigned any NRIS costs.
- NRIS levels can be reduced by any amount during the decision point windows (They just can't be increased).
- If the "Deliv. Level" column is calculated to a negative number then this would mean the project currently is 0% deliverable and one or more Network Upgrades would be required (depending on how many constraints the project is contributing towards) in order to increase the Deliverability.
- **Shared Deduction** (Listed in **Appendix D**) = The shared deduction is the amount by which each study generator affecting a given flowgate/constraint must be scaled down in order to prevent loading violations on the flowgate/constraint. The deduction is proportionally taken from each study generator affecting a particular flowgate/constraint. In summary, it's a reference to how much a project would need to move down by to eliminate a constraint assuming all other projects tagged to that same flowgate/constraint also move down proportionally.

- **Deliverable Level** = “NRIS Requested Amount (MW)” minus “Shared Deduction”.
A project's Deliverable Level would be a project's theoretical maximum NR value that does not contribute to any overload. However, this assumes that ALL other offending projects have reduced their NR requests by the shared deduction amount for each constraint. If this number is negative, it means that there is no delivery level (or NR value) that is possible (given the context of what delivery level references) for the project in question. Only if other study generators drop out of the study process or reduced their NR levels below their shared deduction levels would a project's deliverability level potentially improve.

10. Conditions to GIA (Appendix A10)

The Facilities Study report identifies the cost and schedule of Network Upgrades that are identified for Interconnection projects. In addition to these upgrades, MISO may identify other conditions which may include other higher or similarly queued IRs, other MTEP assumptions embedded in the study case, Distribution Upgrades, or System Protection Upgrades for higher or similarly queued projects.

i. Exhibit A10: MISO will perform analysis on the GI study case and monitor upcoming MTEP upgrades that are not yet in service based on the following Criteria:

- a. $DF \geq 5\%$ AND
- b. MW Impact ≥ 5 MW, AND
- c. MW Impact $\geq 1\%$ of the Facility Rating

Projects that were flagged on Approved MTEP A projects will be treated as conditional and will be subject to MISO's Quarterly/Seasonal/Annual Operating Limit studies. Once the MTEP projects (That study projects were flagged on) have been placed in service, then a project can remove its conditional status.

Study Projects that were flagged on the A10 Analysis:

J1204, J1208, J1209, J1216, J1231, J1232, J1241, J1263, J1266, J1289, J1295, J1299, J1302, J1303, J1306, J1308, J1311, J1332, J1360, J1378, J1382, J1383, J1422, J1447, J1448, J1450, J1453, J1464, J1466

Refer to **Appendix K - A10 Results (CEII)** for detailed breakdowns of each project's A10 results.

11. Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

11.1. Cost Assumptions for Network Upgrades

The cost estimate for each network upgrade identified in System Impact Study was provided by the corresponding transmission owning company.

11.2. Cost Allocation Methodology

The costs of Network Upgrades (NU) for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Study Case.

Cost Allocation Methodology for Thermal Constraints

1. With all Study Group generation projects dispatched in the Post Case, all thermal constraints are identified.
2. Distribution factor from each project on each constraint is obtained.

3. For each thermal constraint, the maximum MW contribution (increasing flow) from each project is then calculated in the Post Case without any network upgrades.
4. For each thermal constraint, the cost estimates for one or a subset of NU are provided by the corresponding Transmission Owner.
5. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each project on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of one NU is:

$$CCPP_{ccPP} PP_{oo} NNNN = \frac{PPPPPPPPPPPP AA PP_{ccPP} PP_{oo} NNNN}{\sum_{i=1}^n \frac{MMNNNN (PPPPPPPP, AA MMMM PP_{ccPP} PP_{oo} NNNN)}{PPPPPPPPPPPP PP_{ccPP} PP_{oo} NNNN}}$$

6. The total NU costs for each project are calculated if more than one NU is required.



Appendix A – Ameren System Impact Study Report (CEII)

Appendix B – Cost Allocation Summary (CEII)

Appendix C – MISO ERIS Analysis (CEII)

Appendix D – MISO Deliverability Analysis (CEII)

Appendix E – Network Upgrades Per Project (CEII)

Appendix F – Ameren LPC Results (CEII)

Appendix G – Ameren Stability Analysis (CEII)

Appendix H – MISO Stability Analysis (CEII)

Appendix I – Short Circuit Studies (CEII)

Appendix J – Affected System Studies (CEII)

Appendix K – Affected System Studies (CEII)